

MA 515: Applied Mathematical Programming

Course Code	MA 515
Course Name	Applied Mathematical Programming
L-T-P-C	3-1-0-4
Prerequisites	NA
Intended for	M.Sc /M.S./Ph.D. /B.Tech 3rd & 4th year
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Origin and types of Linear Program** [5 Lectures]
Model formulation in Industrial Problems, Solution by Graphical Method.
- **Module II** [8 Lectures]
Theory and geometry of linear programs, Simplex Method, Big-M, Two Phase and Revised simplex method, complexity of simplex method, application to decision making.
- **Module III** [7 Lectures]
Duality theory and application, Economic interpretation of dual variables, Primal dual relationship and theorems, Dual simplex method, primal-dual method.
- **Module IV** [5 Lectures]
Integer programming and Applications, Gomory's Algorithm and branch and bound methods.
- **Module V** [5 Lectures]
Transportation problems, Assignment problems, Application in various domains.
- **Module VI** [5 Lectures]
Alternate approaches to solve LPP: ellipsoid method, Karmarkar's algorithm and application.
- **Module VII** [7 Lectures]
Nonlinear programming, Lagrange multipliers, Farka's lemma, constraint qualification, KKT optimality conditions, sufficiency of KKT under convexity; Quadratic Programming, Wolfe method, Industrial Applications of QPP: Machine Learning, Finance etc.

Text Books

1. D. Bertsimas and J. N. Tsitsiklis, **Introduction to Linear Optimization**, Athena Scientific, 1997.
2. Robert J. Vanderbei, **Linear Programming: Foundations and Extensions**, 4th Edition, Springer, 2014.
3. G. V. Reklaitis, A. Ravindran, K. M. Ragsdell, **Engineering Optimization: Methods and Applications**, Wiley, 2006.
4. Mokhtar S. Bazaraa, Hanif D. Sherali and M.C. Shetty, **Nonlinear Programming, Theory and Algorithms**, John Wiley & Sons, 2004.

References

1. Murty, Katta G., ed., **Case Studies in Operations Research: Applications of Optimal Decision Making**. Vol. 212. Springer, 2014.
2. Don T. Phillips, A. Ravindran, James J. Solberg, **Operations Research: Principles and Practice**, John Wiley & Sons, 1987.
3. S. S. Rao, **Engineering Optimization: Theory and Practice**, 4th Edition, John Wiley & Sons, 2009.